ic_autolab round 23 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:54:07Z at git `3f80e412cef62c837fad7408d1f3dea3ad03ef09`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-cf671c` (`python3 tools/pcc/pcc.py standing PCC-20261104-cf671c`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28 **improvement**	wsum2-b60-n2	0.849 [0.782, 0.922]	0.840 [0.768, 0.919]	0.800 [0.705, 0.908]	sel < inc	2.576 / 2.793 / 4.746	3.034 / 3.324 / 5.934	0.934 / 1.021 / 1.804	2.8x / 2.7x / 2.6x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

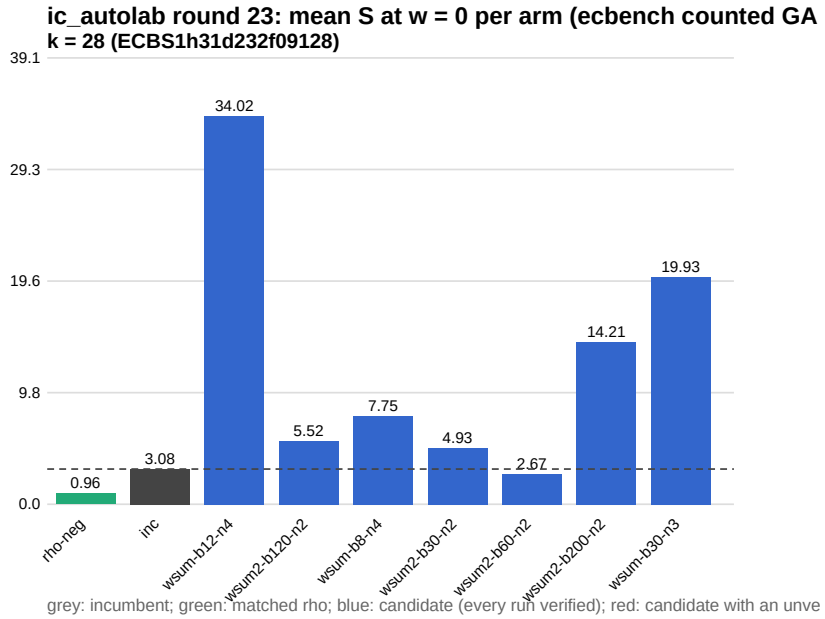
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.8491 [0.782, 0.922] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 28: session `ECBS1h31d232f09128` (run exit 0, verify exit 0, receipt sha256 `6cf7635e0645899e5c69a1687794429f6743e1ece75f325641b779e7620a0654`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	0.961 / 1.051 / 1.857	matched reference
`inc`	wsum-b12	3 `prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=12,free=1`	32	32	3.083 / 3.380 / 6.062	incumbent
`wsum-b12-n4`	wsum-b12	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=12,free=1`	32	32	34.018 / 34.033 / 34.168	incumbent neighbourhood: N = 1.2 x 3
`wsum2-b120-n2`	wsum2-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	5.520 / 5.577 / 6.091	cross-size transfer: wsum2-b120-n2 is the k = 32 incumbent at N / N_a = 0.004
`wsum-b8-n4`	wsum-b8	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=8,free=1`	32	32	7.746 / 7.808 / 8.363	cross-size transfer: wsum-b8-n3 is the k = 24 incumbent at N / N_a = 0.037
`wsum2-b30-n2`	wsum2-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	4.929 / 5.784 / 13.484	exploration: family wsum2-b30 untried at k = 28, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2-b60-n2	wsum2-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.670 / 2.904 / 5.016	exploration: family wsum2-b60 untried at k = 28, N = 0.0001 x N_a
wsum2-b200-n2	wsum2-b200	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=200,free=1	32	32	14.212 / 14.234 / 14.436	exploration: family wsum2-b200 untried at k = 28, N = 0.0001 x N_a
wsum-b30-n3	wsum-b30	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=30,free=1	32	32	19.931 / 19.954 / 20.162	exploration: family wsum-b30 untried at k = 28, N = 0.0001 x N_a



train screen

Selection

- k = 28: challenger `wsum2-b60-n2` (Pareto alternative at w = 1: `wsum2-b60-n2`)

Holdout sessions

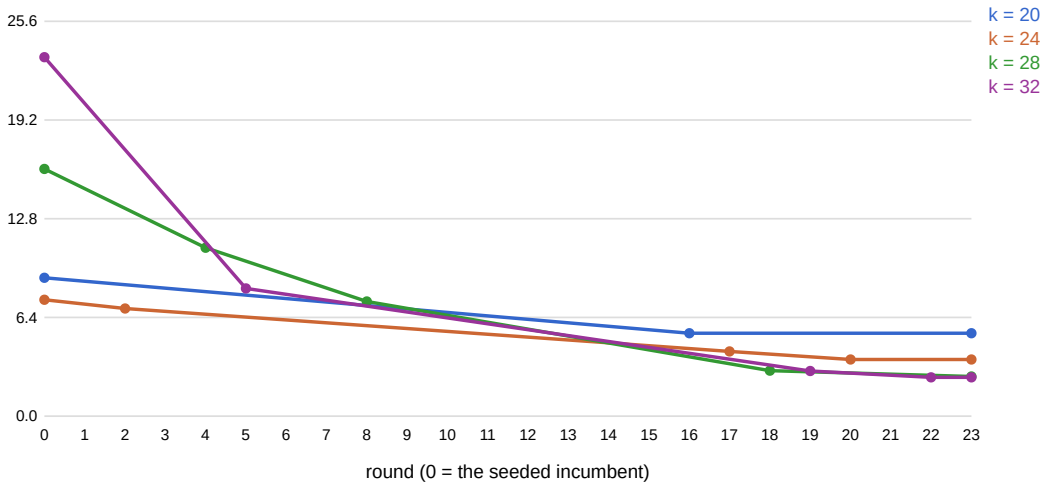
k	session	run exit	verify exit	receipt sha256	levels
28	`ECBS1hf1bfaa6be63c`	0	0	`fe5334c24eb6a3f9736667916e36187eb035381ce8d3a1974efc2d80a844abce`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
24	`wsum-b8-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=8,free=1`)	20	3.678 / 3.920 / 6.100	3.0x / 3.0x / 2.7x
28	`wsum2-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	23	2.576 / 2.793 / 4.746	2.8x / 2.7x / 2.6x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x

Ratchet: incumbent mean S at w = 0 by round (generic track, Curve25519 miniatures)



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.